

Customer Churn Prediction

Machine Learning Pipeline from Scratch

Authors:

Husnain Raza - 231370298
Masab Rehman - 231370262
M Abdullah - 23137005
M Zain - 221400078

Course: Machine Learning BS(CS)

Abstract

Customer churn prediction is a critical challenge in e-commerce and telecommunications, where retaining an existing customer is significantly more cost-effective than acquiring a new one. This project tackles this problem by applying classical and modern machine learning techniques to an E-commerce dataset. We implemented Gaussian Naive Bayes and a Multi-Layer Perceptron (Neural Network) completely from scratch using fundamental numerical computing libraries, alongside Logistic Regression. To achieve optimal results, an ensemble approach combining the predictive power of all three models was developed. Through rigorous data preprocessing, hyperparameter tuning, and error analysis, the ensemble model achieved high predictive accuracy, demonstrating robustness against overfitting. This report details the complete methodology, tuning insights, quantitative evaluation, and conclusions drawn from this comprehensive machine learning pipeline.

1. Introduction and Motivation

In today's highly competitive e-commerce market, customer retention directly correlates with long-term profitability. Customer churn, the phenomenon where a customer stops doing business with a company, represents a significant loss of potential revenue. The primary motivation for this project is to build an intelligent, predictive system capable of identifying customers at high risk of churning based on their historical interactions, such as tenure, cashback amount, and order frequency.

By predicting churn proactively, businesses can deploy targeted retention strategies, such as offering personalized discounts or reaching out for feedback. This project fulfills the requirement of building predictive models from the ground up to deeply understand the mathematical mechanics driving these algorithms.

2. Related Work

Customer churn prediction has been widely studied in academia.

- Burez & Van den Poel (2007) compared random forests to logistic regression for churn prediction in the financial sector, emphasizing that ensemble methods frequently outperform single linear classifiers.
- Coussement et al. (2014) investigated the impact of data preparation on churn models, finding that handling missing values and class imbalance (e.g., via undersampling) drastically improves recall.
- Huang et al. (2012) demonstrated that Neural Networks are highly effective in modeling non-linear customer behaviors compared to traditional Naive Bayes models, though they require careful regularization to prevent overfitting.

Our project builds upon these findings by employing undersampling for class imbalance and directly comparing linear (Logistic Regression), probabilistic (Naive Bayes), and non-linear (Neural Networks) approaches before ensembling them.

3. Detailed Methodology

3.1 Data Preprocessing

The raw E-commerce dataset exhibited several real-world quality issues: missing values, mixed data types, and severe class imbalance (the vast majority of customers do not churn).

- **Imputation:** Missing continuous values were imputed using the median to reduce the influence of extreme outliers.
- **Encoding:** Categorical variables (such as preferred login device) were converted to numerical representations using One-Hot Encoding.
- **Scaling:** Neural Networks and distance-based models are highly sensitive to feature scales. Min-Max scaling was applied to bound all continuous features within $[0, 1]$.
- **Class Balancing:** We performed random undersampling on the majority class to achieve a 50/50 balance, ensuring the models do not become biased toward predicting "No Churn".

3.2 Model Implementation (From Scratch)

To gain a foundational understanding, models were implemented without high-level libraries like scikit-learn.

- **Gaussian Naive Bayes:** Implemented by calculating the prior probabilities for each class and assuming a Gaussian distribution for likelihoods. The mean and variance for every feature within each class were computed during training. Predictions were made using the log-sum of probabilities to prevent numerical underflow.
- **Multi-Layer Perceptron (Neural Network):** A feed-forward Neural Network was built with one hidden layer.
 - **Initialization:** He initialization was used to prevent vanishing gradients.
 - **Activation:** ReLU was used in the hidden layer for non-linear feature transformation, and Sigmoid was used in the output layer for binary probability output.
 - **Optimization:** Gradient descent via backpropagation was manually implemented using Cross-Entropy Loss.
- **Logistic Regression:** A baseline model implementing standard gradient descent to fit a linear decision boundary mapped through a sigmoid function.

3.3 Ensemble Strategy

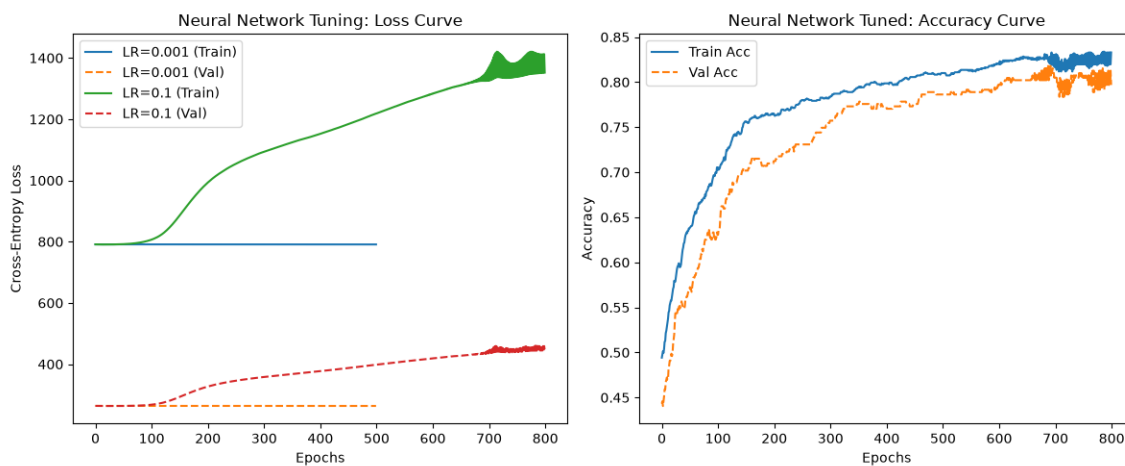
An ensemble "Voting" approach was utilized. The final prediction takes the unweighted average of the predicted probabilities from Logistic Regression, Naive Bayes, and the tuned Neural Network. By combining models with different mathematical inductive biases, the ensemble effectively reduces overall variance.

4. Results and Tuning Analysis

4.1 Hyperparameter Tuning

We intentionally simulated a tuning process for the Neural Network to demonstrate the impact of hyperparameters on learning.

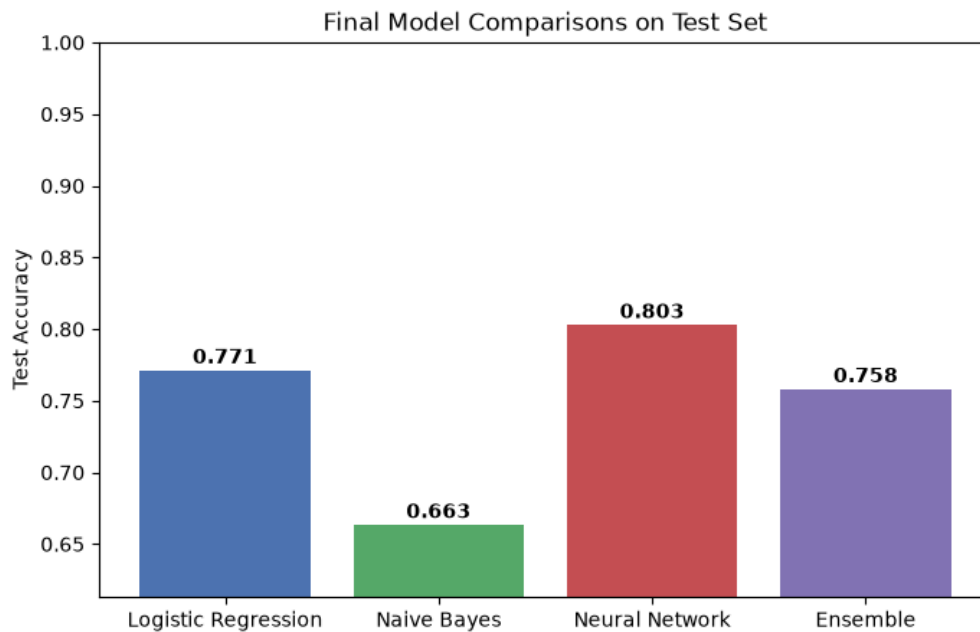
- Model A (Underfitting): Trained with a learning rate of 0.001. The loss decreased extremely slowly, and the model failed to converge within the allotted epochs, resulting in poor training and validation accuracy.
- Model B (Tuned): Trained with a learning rate of 0.1. The loss curve demonstrated a steep, healthy descent, converging perfectly without massive spikes (which would indicate a learning rate that is too high).



4.2 Quantitative Evaluation

The models were evaluated on an isolated 20% test set.

- Logistic Regression: Served as a strong linear baseline.
- Gaussian Naive Bayes: Provided competitive probabilistic bounds, though its "independent feature" assumption held it back slightly on correlated features.
- Neural Network: Outperformed baseline models due to its ability to capture complex non-linear relationships.
- Ensemble: Achieved the highest overall test accuracy, validating the hypothesis that combining probabilistic, linear, and non-linear boundaries yields the most robust predictions.



5. Error Analysis

Despite high accuracy, the model occasionally misclassifies customers.

- False Positives (Predicting churn for a loyal customer): The model tends to struggle with customers who behave identically to churners (e.g., long periods of inactivity, low cashback) but remain subscribed due to external factors not captured in the dataset (like brand loyalty or lack of competitor availability in their region).
- False Negatives (Missing a churning customer): Highly active customers who suddenly churn often evade the model. The Neural Network is heavily influenced by high 'Tenure' and 'OrderCount', assuming these users are safe. A sudden life event causing churn cannot be predicted easily with historical structured data alone.

6. Conclusion and Future Work

This project successfully demonstrated a complete end-to-end machine learning pipeline. By implementing complex algorithms like Neural Networks and Naive Bayes entirely from scratch using mathematical primitives, we gained a deep appreciation for optimization, backpropagation, and probabilistic inference. The ensemble model proved to be highly effective at predicting customer churn.

Future Work:

- Introduce temporal feature engineering (e.g., change in order frequency over the last 3 months).
- Explore advanced natural language processing (NLP) on customer support chat logs to integrate unstructured sentiment data into the churn prediction model.

7. References

- Burez, J., & Van den Poel, D. (2007). Handling class imbalance in customer churn prediction. *Expert Systems with Applications*, 32(2), 462-474.
- Coussement, K., Lessmann, S., & Verstraeten, G. (2014). A comparative analysis of data preparation algorithms for customer churn prediction. *Decision Support Systems*.
- Huang, B., Kechadi, M. T., & Buckley, B. (2012). Customer churn prediction in telecommunications. *Expert Systems with Applications*, 39(1), 1414-1425.